AKSHAY KHULE

Principal Architect | Distributed Systems · Data Platforms · Generative AI

Mobile: 9326768815 | Email: Khule.akshay@gmail.com

LinkedIn | **GitHub** | **Cloud Certifications** | **Distributed Systems Certifications**

PROFESSIONAL SUMMARY

16+ years of making "seemingly complex" distributed systems simple.

- 16+ years of global technical leadership across Software Engineering, Product Development, and Systems Architect - architecting and shipping distributed systems, cloud-native microservices, terabyte-scale data platforms, and GenAI solutions across Banking, Pharma, AIOps, and Cybersecurity domain.
- Architecture leadership across multiple product teams (led teams of 30+); delivered winning RFP responses for new business; accountable for technical design, product integration, and compliance with architecture and design principles.
- Hands-on product architecture and development of microservices and serverless applications — Java/J2EE, Spring Boot, Spring Cloud, Spring Security, OAuth/KeyCloak, JPA, REST, RDBMS with database schema design, and message queues — engineered with configuration management, service discovery and registration, resiliency patterns, and distributed log tracing, containerized with Docker and deployed on Kubernetes via Helm across AWS and Azure.
- Expert in end-to-end Big Data and Data Lake solutions — streaming and batch ingestion, storage, processing, consumption, and governance with cross-cutting concerns — using Spark (Databricks, Spark Core, Spark SQL), Kafka and Kafka Streams (producers and consumers), Delta Lake, Azure Synapse, Azure Data Factory, Microsoft Fabric, and cloud storage — delivering terabyte-scale pipelines with zero-data-loss guarantees.
- Hired, mentored, and guided Application Development and Data Platform teams (architects, lead developers, engineers) — driving code reviews and engineering best practices, evaluating emerging open-source technologies through POCs, and encouraging cross-team collaboration. Defining and implementing the overall architecture strategy to align IT with business.
- Generative AI: multi-agent orchestration (Agno framework), RAG, and AI guardrails on AWS Bedrock and Microsoft Foundry, applied to incident triage and operations automation.

PUBLICATIONS & THOUGHT LEADERSHIP

- **White paper:** GenAI-Augmented Polyglot Persistence for Microservices — Toward Self-Adaptive & Explainable Data Fabrics. **https://lnkd.in/dVeegA4H**
- **Newsletter:** The Engineering Log by Akshay — ongoing LinkedIn publication on distributed systems, data engineering, and GenAI. **Read on LinkedIn**

CERTIFICATIONS

AWS	Solutions Architect Professional | Data Analytics Specialty | Security Specialty
Databricks	Databricks Certified Professional | Spark Certified Associate Developer
Confluent (Kafka)	Confluent Certified Developer for Apache Kafka
Microsoft Azure	Microsoft Certified: Azure Fundamentals
DAMA	Certified Data Management Professional (CDMP)
IIT Roorkee	Generative AI Engineering Program (completed)
Anthropic	Claude 101 — Certificate of Completion

TECHNICAL SKILLS

Languages	Java, Python, Scala, SQL, PL/SQL
Big Data & Analytics	Apache Spark (Databricks, Spark Core, Spark SQL), Apache Kafka, Kafka Streams, Delta Lake, Hive, HDFS, Hadoop, EMR, Azure Synapse Analytics, Azure Data Factory, Microsoft Fabric (OneLake), AWS Glue, Redis, Elasticsearch, Data Governance
Cloud & Containers	AWS, Azure, Kubernetes (AKS), Docker, Helm, Databricks Asset Bundles, GitHub Actions, Azure Logic Apps, Azure Functions, AWS Lambda, Synapse Dedicated pool
Frameworks & Tools	Microservices, Spring Boot, Spring Cloud, Spring MVC, Spring Security, Flask, Hibernate ORM, EJB, JMS, MDB, REST, OAuth, KeyCloak, ProtoBuff , JWT, JUnit, Jenkins, Maven, Git, SonarQube, Splunk
Generative AI	LLMs (GPT-4o, Claude), Agentic AI, Multi-Agent Orchestration (Agno), RAG, AI Guardrails, AWS Bedrock, Microsoft Foundry

PROFESSIONAL EXPERIENCE

Principal Architect — Unisys Corporation *May 2021 – Present*

Architect (2021 – 2024); Principal Architect (2024 – Present)

Unisys CloudForte - Enterprise Monitoring & Security Product (Unisys Stealth Family)

Observability product in the Unisys Stealth cybersecurity portfolio - monitors enterprise infrastructure and applications in real time, providing the telemetry foundation for anomaly detection, security monitoring, and threat visibility.

URL: **marketplace.microsoft.com — CloudForte for Azure**

- Designed the overall product architecture, cloud platform setup, and technology choices for CloudForte. Built the event-processing layer as Spring Boot microservices on Azure Kubernetes Service (AKS) - consuming ~5M events/day safely (no duplicates, automatic retries, dead-letter queues for failed messages) and triggering alerts or automated fixes based on risk
- Designed a GenAI incident-triage agent (Agno framework) that recommends fixes from runbooks and past incidents, drafts incident summaries, and estimates impact - designed so the AI only recommends, while actual fixes execute through approved, tested runbooks. Achieved 30% MTTR reduction.

Banking Data Platform Modernization — Financial Services Client

- Designed the migration strategy after Microsoft deprecated the Export to Data Lake service — built a data-ingestion solution pulling Dynamics 365 F&O data from Microsoft Fabric (Fabric Link/OneLake) into an Azure Synapse dedicated SQL pool, with zero disruption to downstream financial reporting across ~100+ pipelines.
- Developed a PySpark-based solution in Azure Databricks (Delta Lake) to process Dynamics 365 F&O financial data, migrating ledger and financial-transaction processing into scalable, version-controlled Spark workloads — cutting pipeline runtime from ~3 hrs to 20 mins. Implemented Databricks Asset Bundles (DABs) with GitHub Actions CI/CD to declaratively define, promote, and gate deployments across dev, staging, and production environments.

Architect — Cognizant (Client: Novartis) *July 2018 – May 2021*

Adverse Event resolution product for Novartis — analyzes adverse events and predicts resolutions.

URL: https://ae-service.novartis.net/pae-coreui

- Designed and developed the full system — Salesforce Social Studio Stream, microservices, AWS Lambda, API Gateway, and SQS communicating over REST — optimizing operations through cloud adoption and saving nearly $1M in annual costs.
- Led the build-out of the Strategic Data Platform, redefining data processing, archival, and retrieval for the data-ingestion product — solving data lineage, latency, stewardship, and governance problems. Uplifted the stack from synchronous to asynchronous (SQS + Lambda); designed the big-data skeleton consolidating RDBMS and file-system sources into an S3 data lake for downstream analytics; hands-on coding on critical modules.

Technical Manager — Capgemini (Client: Morgan Stanley) *Feb 2017 – June 2018*

- Owned end-to-end product development — architecture, design, development, release, and maintenance of Spring Boot microservices: defined service boundaries and API contracts, implemented REST controllers, service and DAO layers, and designed entities and the underlying data model.
- Designed the microservices architecture with clear separation of concerns — stateless services, layered design (controller/service/repository), centralized configuration, and inter-service communication over REST - enabling independent development, testing, and release of services.

Technical Lead — Persistent Systems (Client: Thermo Fisher) *Sept 2014 – Jan 2017*

Next-Generation Sequencing product for Thermo Fisher — automates sample/library preparation, sequencing, analysis, and reporting. ***URL:*** ***https://ionreporter.thermofisher.com***

- Implemented REST controllers, service and DAO layers, and entity definitions; built the persistence layer with Hibernate.

System Analyst — Worldline (Client: BNP Paribas) *Apr 2013 – Aug 2014*

Worldline SEPA solutions — an all-in-one payment ecosystem that automates, secures, and streamlines Single Euro Payments Area (SEPA) transactions across Europe.

- Built a reusable Spring/Spring REST transaction engine and Hibernate ORM data layers for liquidity and transaction management — optimizing SEPA direct-debit and credit-transfer processing.

Consultant — Polaris (Client: JPMorgan) *Mar 2011 – Mar 2013*

TLIP banking application for JP Morgan — integration of the TIDE, LIQUIDITY, and PORTAL modules. ***URL:*** ***intellectdesign.com***

- Designed core subsystem boundaries, application interfaces, and multi-tier enterprise runtime platforms using EJB and JMS; collaborated with QA on complex test analysis, defect resolution, and performance validation.

Consultant — HAL (in-contract) *Jan 2009 – Mar 2011*

- Building an in-house inventory management and supply chain solution with batch application support for an ERP.

KEY ENGINEERING HIGHLIGHTS

Distributed Systems Design

- High-Throughput Partitioning: diagnosed and fixed hot-partition skew in high-velocity Kafka pipelines by redesigning topic partition keys (entity-level composite keys) and migrating consumers to cooperative-sticky rebalancing.
- Fault-Tolerant Design: designed cloud data-ingestion services for partition tolerance — idempotent, retry-safe operations with exponential backoff and jitter, plus dead-letter queues.

Massive-Scale Data Ingestion & Engineering

- Terabyte-Scale Ingestion: architected a high-velocity ingestion engine on Spark handling terabytes of records with strict zero-data-loss guarantees. Low-Latency Streaming: implemented event-driven Kafka pipelines with fault-tolerant Java consumer-group services — cutting data availability to minutes.

Microservices & Backend

- Event-Driven Services: built SQS-based microservices using the transactional outbox pattern - saving business data and the outgoing event in one database transaction before publishing — with idempotent consumers safely handling duplicates; removed data inconsistencies between payment and notification services.
- Framework Stabilization: established scalable enterprise backend skeletons with Java, Spring Boot, Spring Cloud, and Hibernate ORM — custom entity scopes, transaction boundaries, and efficient lazy-loading configurations.

DevOps & CI/CD

- Infrastructure as Code: architected Databricks Asset Bundles to declaratively define, version, and promote data workloads across environments — integrated with secure GitHub Actions pipelines on VNet-isolated self-hosted runners with strict network boundaries, environment-specific variable injection, and gated approvals.
- Containerized Orchestration: containerized distributed applications with Docker and engineered clean Helm charts for automated, fault-tolerant orchestration on managed Kubernetes (AKS).

Generative AI

- Multi-Agent Orchestration: built multi-agent systems (Agno) where specialized retrieval, reasoning, and execution agents coordinate task flows, autonomously handling complex workflows end-to-end.

AWARDS & RECOGNITION

- **Unisys All-Hands Recognition:** recognized four times at company-wide All-Hands meetings for high-impact work across distributed systems and data engineering initiatives.
- **"You Rock" Award Nomination — Persistent:** nominated by peers and leadership for consistent technical excellence.

EDUCATION

- Bachelor of Engineering (B.E.), Computer Science — V.Y.W.S. College of Engineering, Badnera, Amravati University (1st Division).
- Generative AI Engineering Program — IIT Roorkee.